

Complex Analysis – 1

Домашнее задание

1 сентября 2026 года

Теоретический минимум. $|z|^2 = z\bar{z}$ and $\bar{z} = 1/z$ on the unit circle; roots of unity and the summation filter; Vieta's formulas; the theorem of Gauss–Lucas on the roots of p' ; the Eneström–Kakeya bound for polynomials with monotone positive coefficients; the maximum principle for polynomials; the geometry of $z \mapsto \omega z$ (rotation) and of the cube root of unity in problems about equilateral triangles.

1. (Putnam 1989) Prove that if $11z^{10} + 10iz^9 + 10iz - 11 = 0$, then $|z| = 1$.
2. (Putnam 1973) Let $z = x + iy$ be a complex number with x and y rational and $|z| = 1$. Show that $|z^{2n} - 1|$ is rational for every integer n .
3. (Putnam 1975) For which ordered pairs of real numbers b, c do both roots of $z^2 + bz + c = 0$ lie inside the unit disc $\{|z| < 1\}$? Draw the corresponding region of the real bc -plane and identify its boundary curves precisely.
4. (Putnam 1947) Let $P(z) = z^2 + az + b$ be a quadratic polynomial with complex coefficients. Suppose that $|P(z)| = 1$ for every z with $|z| = 1$. Prove that $a = b = 0$.
5. (Putnam 1949) How many roots of the equation $z^6 + 6z + 10 = 0$ lie in each quadrant of the complex plane?
6. (Putnam 1957) Let P be a complex polynomial of degree n whose roots can be covered by a closed disc of radius R . Show that the roots of $nP(z) - kP'(z)$ can be covered by a closed disc of radius $R + |k|$, where k is any complex number. (*Подсказка: сначала Гаусс–Люка, потом сдвиг центра.*)

7. (Putnam 1959) Prove that if the points of the complex plane corresponding to two distinct numbers z_1 and z_2 are two vertices of an equilateral triangle, then the third vertex corresponds to $-\omega z_1 - \omega^2 z_2$, where ω is an imaginary cube root of unity.
8. (Высшая лига 2025) Let f be a monic polynomial of degree n . Prove that there exists z with $|f(z)| \leq 1$ and $|f'(z)| \geq n$.
9. (Putnam 1979) For $k = 1, \dots, n$ let $z_k = x_k + iy_k$ with x_k, y_k real, and let r be the absolute value of the real part of $\pm \sqrt{z_1^2 + z_2^2 + \dots + z_n^2}$. Prove that $r \leq |x_1| + |x_2| + \dots + |x_n|$.
10. (Putnam 1972) Let $n_1 < n_2 < \dots < n_k$ be positive integers. Prove that the polynomial $1 + z^{n_1} + z^{n_2} + \dots + z^{n_k}$ has no roots in the disc $|z| < \frac{\sqrt{5}-1}{2}$.